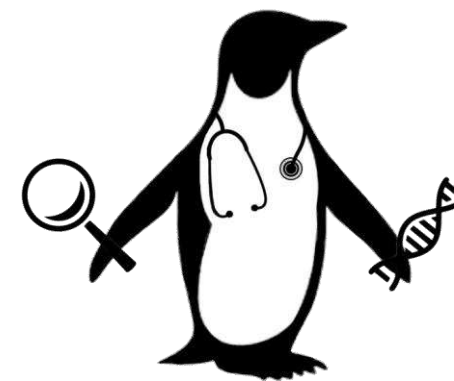




PENGQUIN

# PENGQUIN:

# PErsoNal Genome and QUery in healthcare and clinical practice

**PhD Candidate:** Elias Crum

VITO 18-month jury  
Nov 20th, 2025

**Advisors:** Dr. Gökhan Ertaylan<sup>1</sup>, Dr. Bart Buelens<sup>1</sup>, Dr. Ruben Taelman<sup>2</sup>, Prof. Dr. Ruben Verborgh<sup>2</sup>

1 = 

2 = 

# Presentation Overview

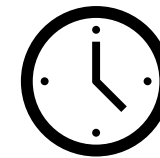
## **Project Review**



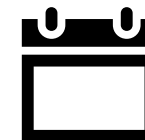
~24 Month Progress



Current Undertakings



Remaining PhD Plans

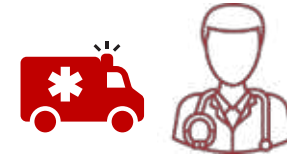


# PENGQUIN:

PErsoNal Genome QUery IN  
healthcare and clinical practice



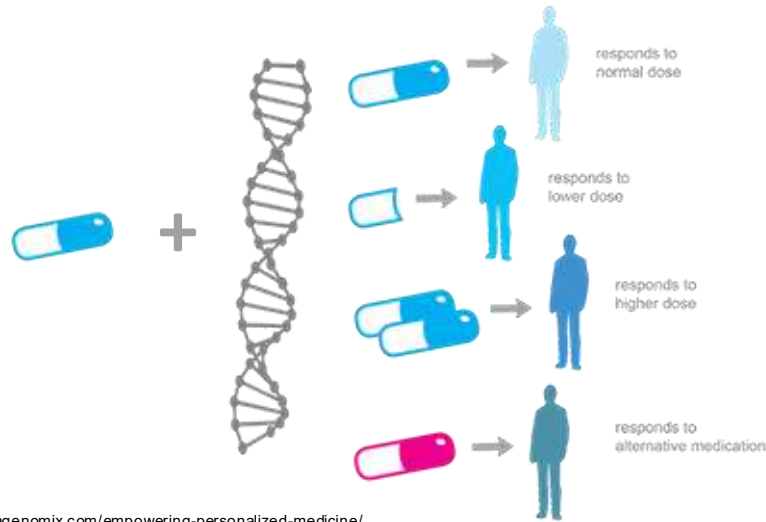
Genomic Data



Clinical Care

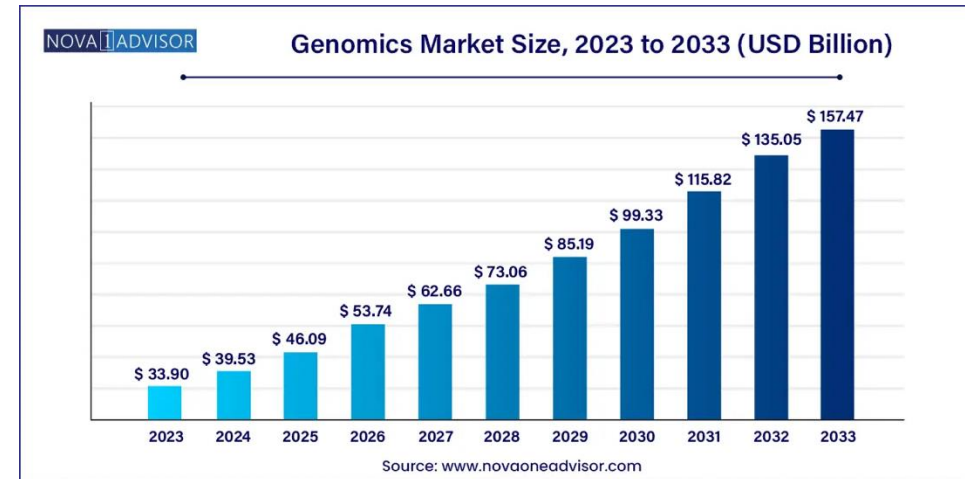
# Genomics + Healthcare

## A Example: Pharmacogenomics



<https://alphagenomix.com/empowering-personalized-medicine/>

## B Genomics market is growing RAPIDLY



<https://www.novaoneadvisor.com/report/genomics-market>

## C

### Institution-centric

#### Data Storage



#### Infrastructure



?

#### Data Usage

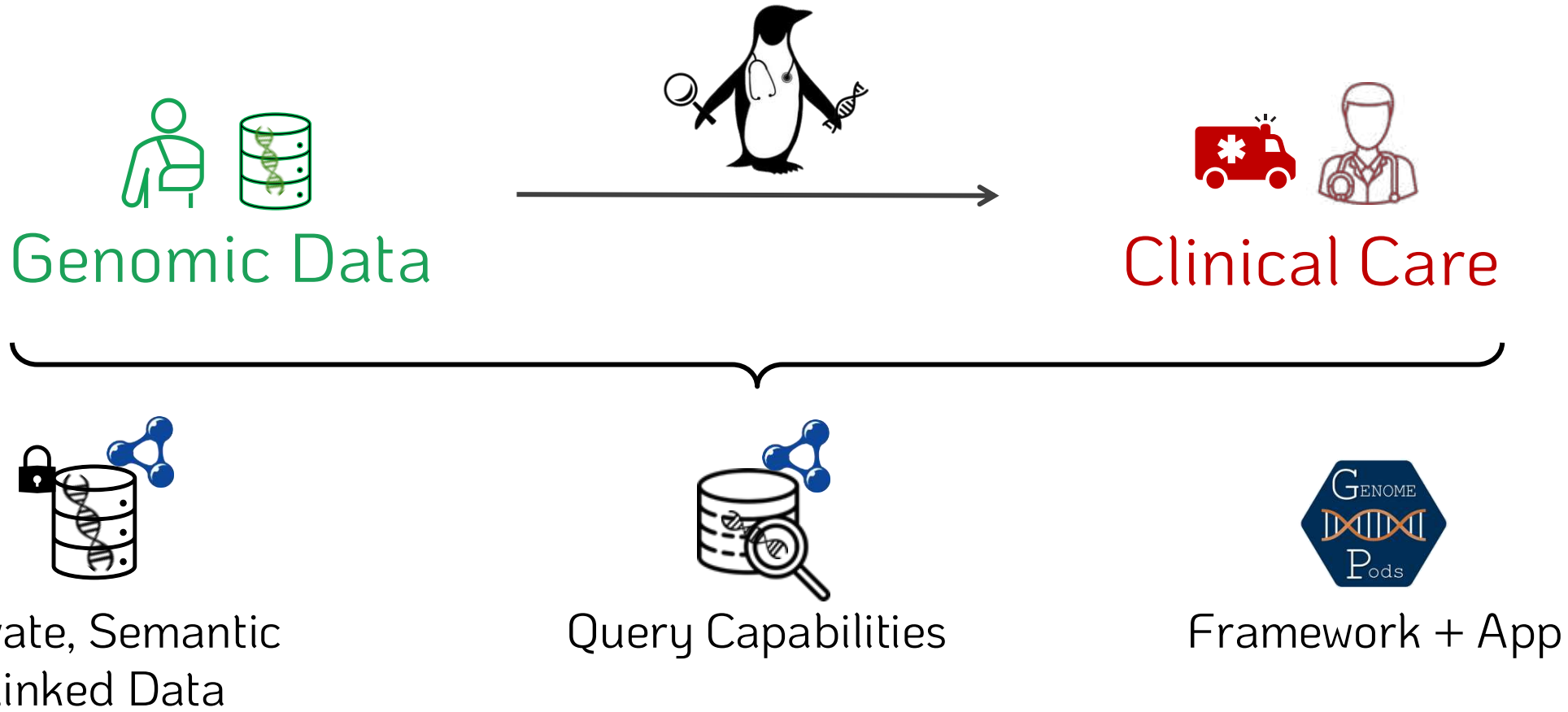


**Low Efficiency  
+  
High Costs**

**Problem:**

# PENGQUIN:

PErsoNal Genome QUery IN  
healthcare and clinical practice



# Genomic Data for Clinical Use

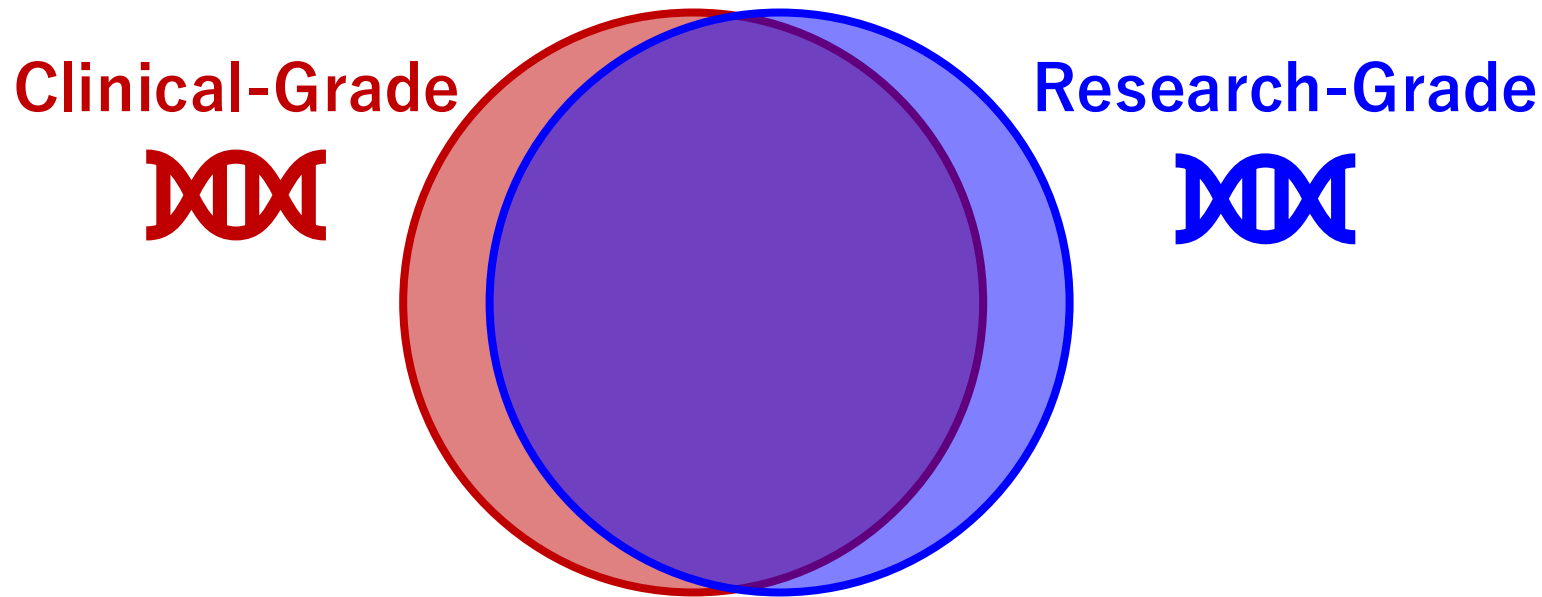
-  Format is use-case specific
-  Whole Genome Sequence (WGS) data
-  Large Projects: 1+ Million Genomes (EU)<sup>1</sup>



<sup>1</sup> <https://digital-strategy.ec.europa.eu/en/policies/1-million-genomes>

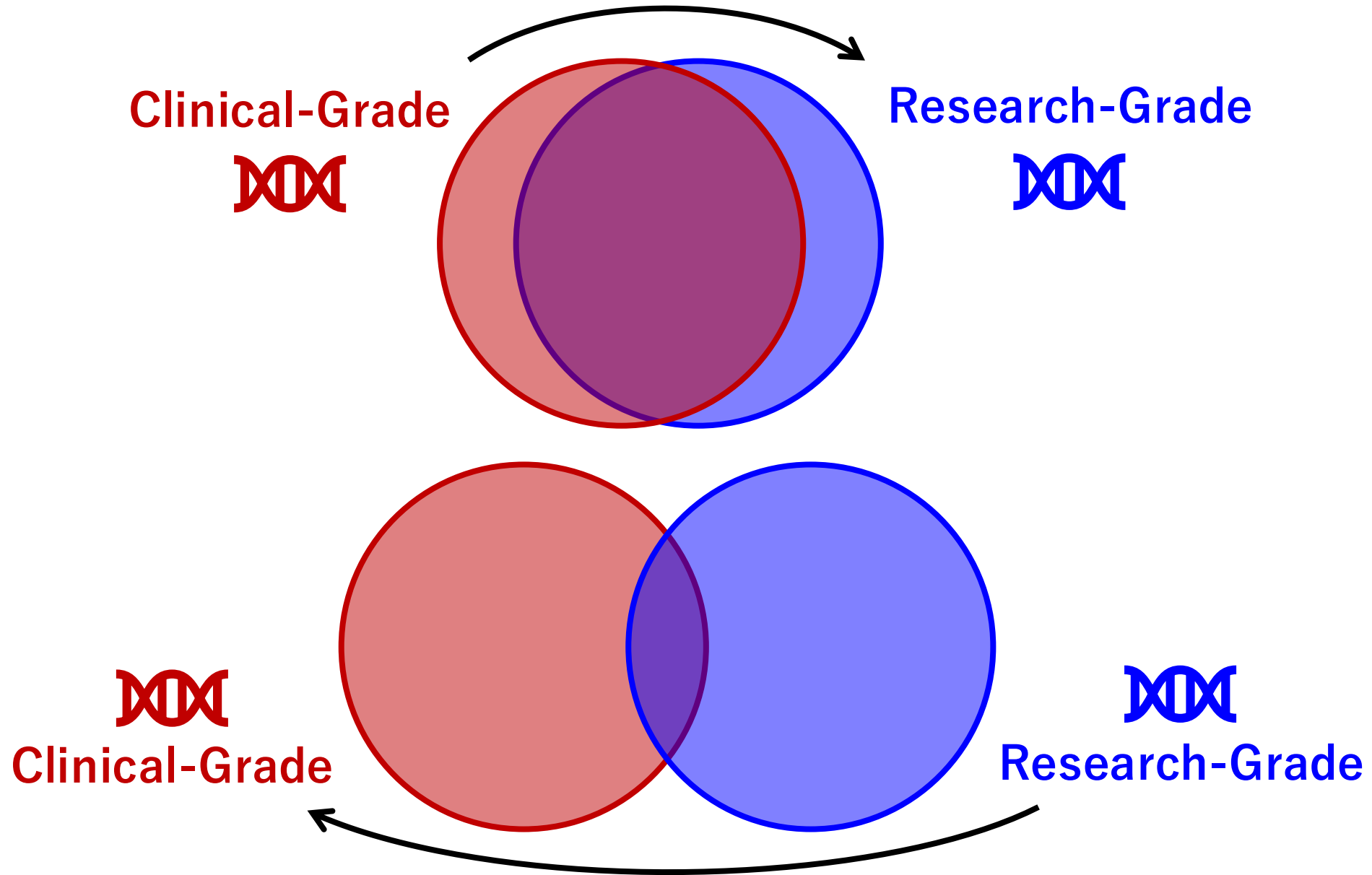
# Genomic Data for Clinical Use (assumption)

-  Format is use-case specific
-  Whole Genome Sequence (WGS) data
-  Large Projects: 1+ Million Genomes (EU)<sup>1</sup>



<sup>1</sup> <https://digital-strategy.ec.europa.eu/en/policies/1-million-genomes>

# Genomic Data for Clinical Use (reality)



# The current landscape

Intended use is VERY important

Clinical use  $\neq$  Research use

**\*\* Funding goes predominantly to RESEARCH-GRADE \*\***

Cheaper

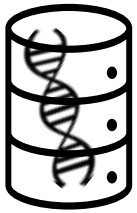
Fewer regulatory challenges

Funders / Policy makers misunderstanding

**\*\* But how do we “translate” to clinical use?? \*\***

# Sharing Genomic Data – Technology Considerations

## Key Implementation Factors



Data Storage

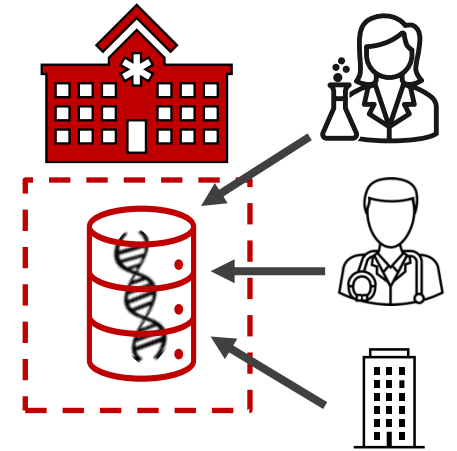
Considerations:

- patient consent
- data security
- access controls

&

Considerations:

- consent changes
- governance
- infrastructure

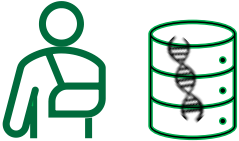


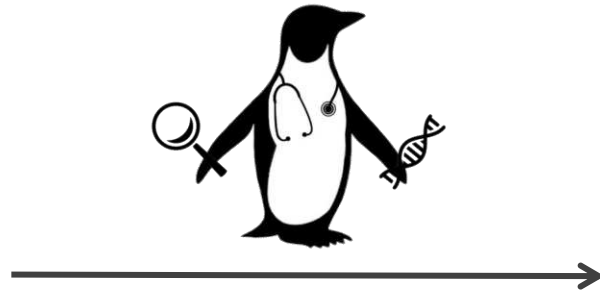
Data Sharing

A more in-depth analysis of these challenges + potential solutions detailed  
in an upcoming publication in *Computers in Biology and Medicine*

# PENGQUIN:

PErsoNal Genome QUery IN  
healthcare and clinical practice

  
Genomic Data



  
Clinical Care



Framework + App

**1. Genomic Data Interaction**



Semantic  
Representation

**2. Semantify Genomic Data**



Query Capabilities

# 1. Data Interaction Framework

*A Web Application to:*

 *Upload Data to a Solid Pod*

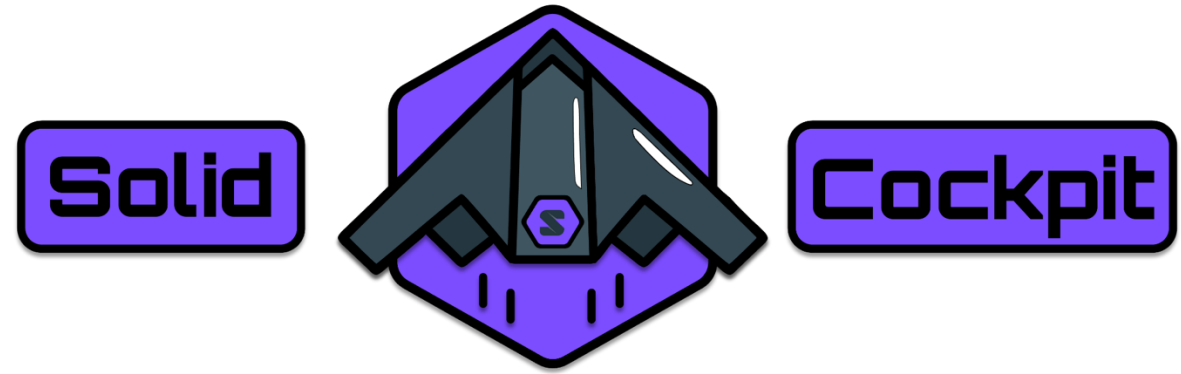
 *Browse contents of a Solid Pod*

 *Edit Privacy of (and share) data in a Solid Pod*

 *Query Data in Solid Pods + SPARQL Endpoints*

 *Includes a query cache for storing queries and results*

*\*\* Enables patient, provider, and auditor control over data \*\**

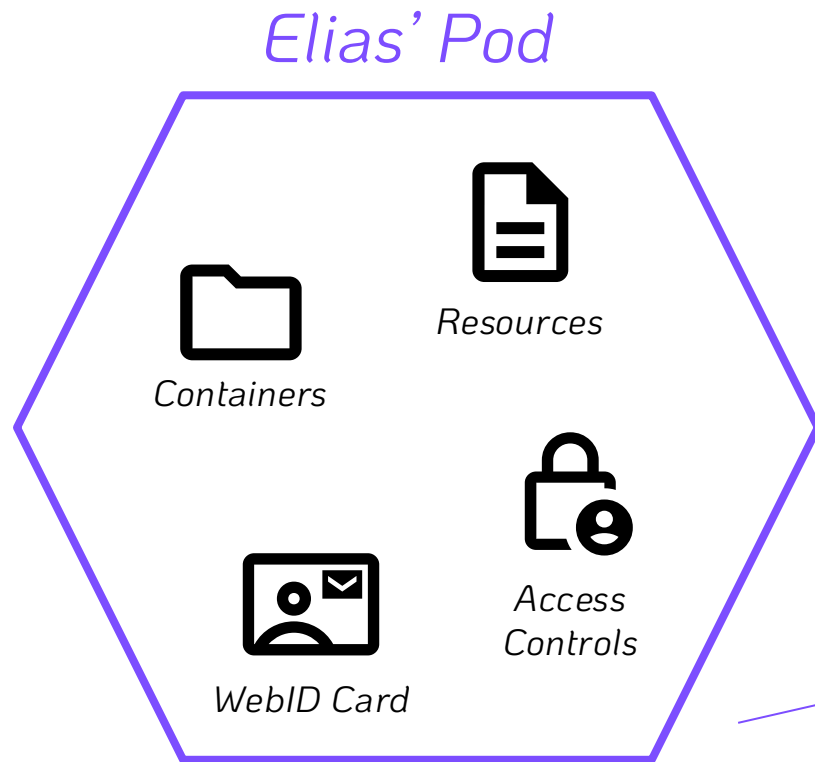


# What is a Solid Pod?



*Secure, Decentralized Data Storage Specification*

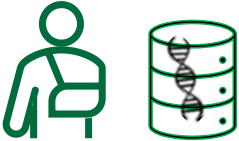
- ↳ *Pod exists on a Server (accessible to the Web)*
- ↳ *Can store files, RDF data, and Linked Data*
- ↳ *Provides control policies for stored data*



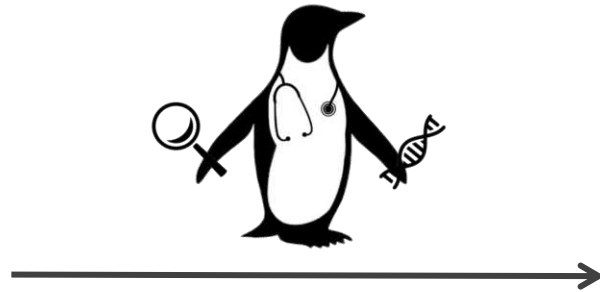
**\*\* Enables secure, authenticated sharing via HTTP protocols \*\***

# PENGQUIN:

PErsoNal Genome QUery IN  
healthcare and clinical practice



Genomic Data



Clinical Care



Framework + App

**1. Genomic Data Interaction**



Semantic  
Representation

**2. Semantify Genomic Data**



Query Capabilities

# 2. Genomic Data Representation

From a file-based approach

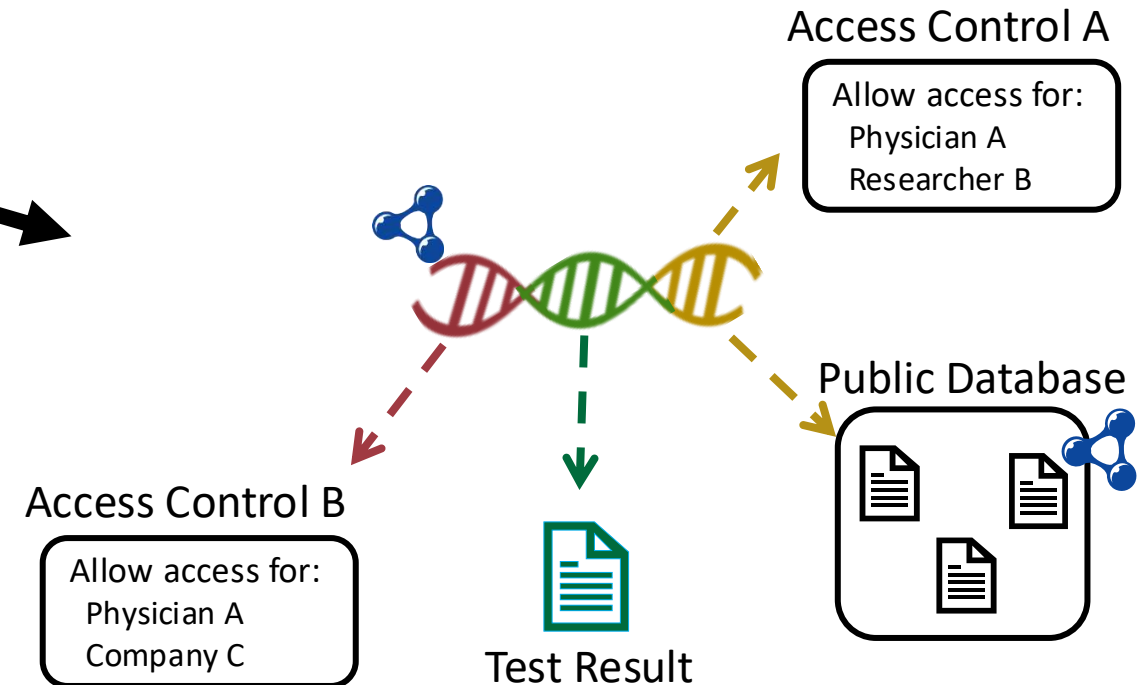


To a granular, dynamic RDF approach

Conversion using existing ontologies

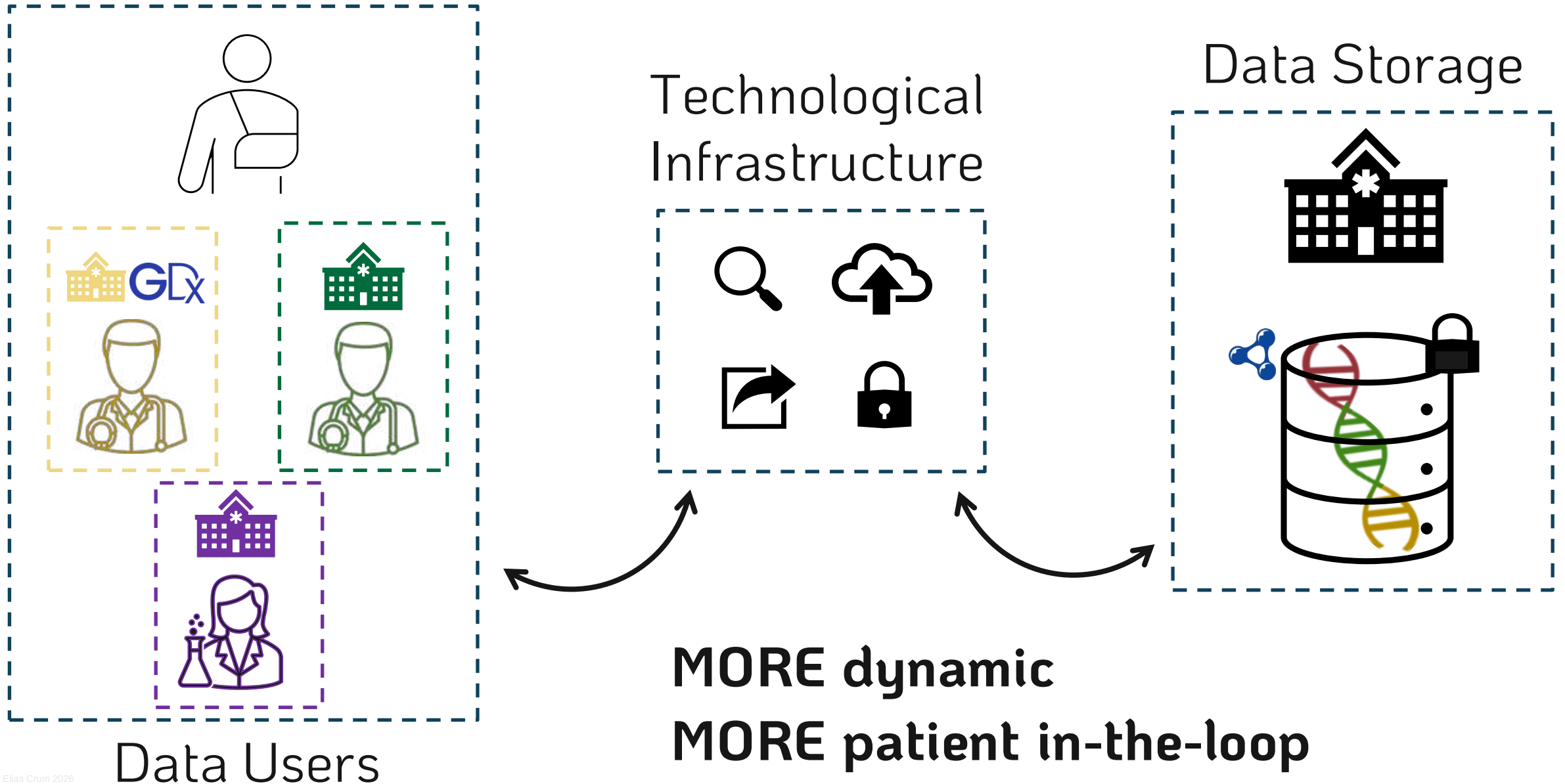
Policy definition via ODRL<sup>1</sup>

Implementation within Solid Pod Env



<sup>1</sup> <https://www.w3.org/TR/odrl-model/>

# Future Possibilities?



# Thank you!



More PhD specifics



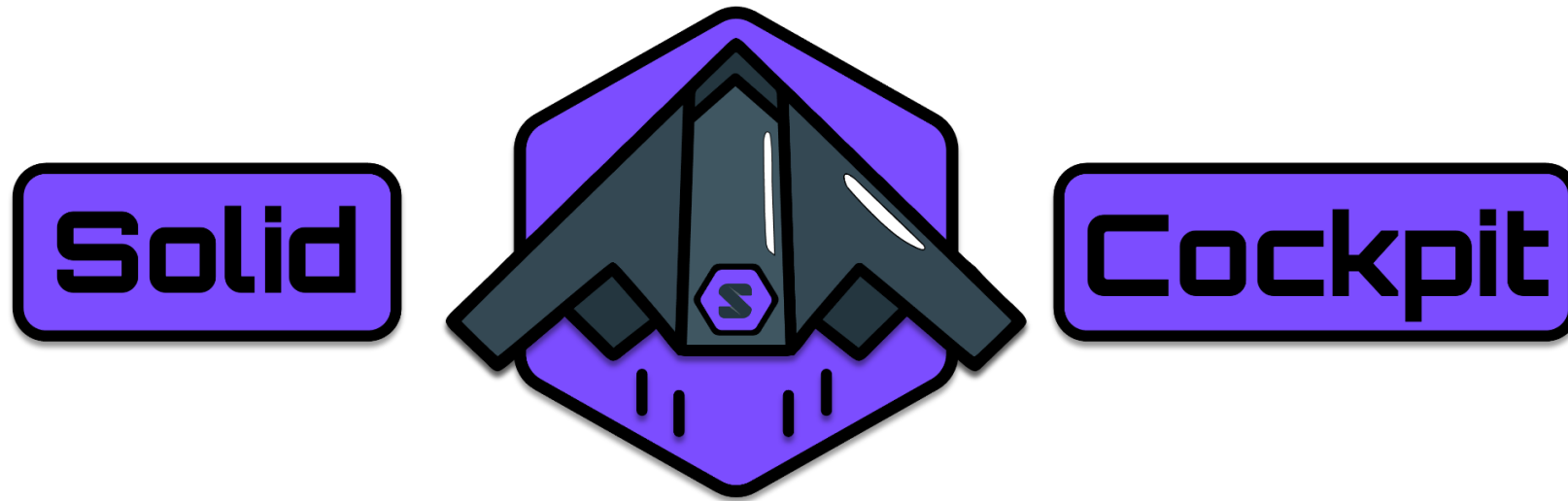
LinkedIn



GitHub

# Informal Demo / Tutorial Later Today

(for those interested)



[knowledgeonwebscale.github.io/solid-cockpit/](https://knowledgeonwebscale.github.io/solid-cockpit/)

# Supplemental Slides

# Genomic Data

## Clinical-Grade

Higher read quality  
More expensive  
Stricter use-regulations

## Research-Grade

Lower read quality  
Less expensive  
Flexible use-regulations

| Dimension                        | Clinical Grade Genome Data Sharing                           | Research Grade Genome Data Sharing                       |
|----------------------------------|--------------------------------------------------------------|----------------------------------------------------------|
| 1. Primary Purpose and Scope     | Patient care and medical decision-making.                    | Discovery and hypothesis-driven studies.                 |
| 2. Data Accuracy and Quality     | Stringent quality control (QC) and validation.               | High but no absolute QC standards.                       |
| 3. Regulatory Requirements       | Subject to medical regulatory frameworks.                    | Subject to research ethics regulations.                  |
| 4. Consent Requirements          | Clinical-informed consent.                                   | Research-informed consent.                               |
| 5. Privacy and Confidentiality   | Highest level of patient privacy required.                   | Strong but variable privacy measures.                    |
| 6. Security Controls             | Robust, enterprise-grade security.                           | Secure but more flexible.                                |
| 7. Data Governance and Oversight | Hospital/clinic-based governance.                            | Institution or consortium-based governance.              |
| 8. Ethical Considerations        | Focus on direct patient benefit and “do no harm.”            | Focus on scientific advancement and participant welfare. |
| 9. Infrastructure Requirements   | Highly reliable, clinically validated systems.               | Flexible, scalable research infrastructure.              |
| 10. Architecture Requirements    | Standardized, interoperable clinical systems.                | Flexible, modular architectures.                         |
| 11. Legal Liabilities            | High.                                                        | Moderate.                                                |
| 12. Interoperability             | High necessity for clinical integration.                     | Variable.                                                |
| 13. Data Ownership and Control   | Patient-oriented ownership with institutional custodianship. | Institution or sponsor-owned.                            |
| 14. Data Sharing Scope           | Limited to necessary clinical stakeholders.                  | Broad or consortium-based.                               |
| 15. Cost and Funding Models      | Healthcare reimbursement or institutional budgets.           | Grant-funded or institutional research budgets.          |

# Genomic Sequence Data Sharing for Clinical Practice: A Scoping Review



**Aim:** Identify existing implementations, guidelines, and challenges of clinical sharing of genomic data.

**Methods:** Reviewed 55 studies, 57 national genomics projects, and 20 major genomics companies.

**Results:** Clinical genome sharing implementations exist, yet scalability and widespread adoption remain limited.

**Suggested steps forward:**

Harmonize legal bases for  
secondary clinical use of  
research data



Mandate (GA4GH)  
technical standards



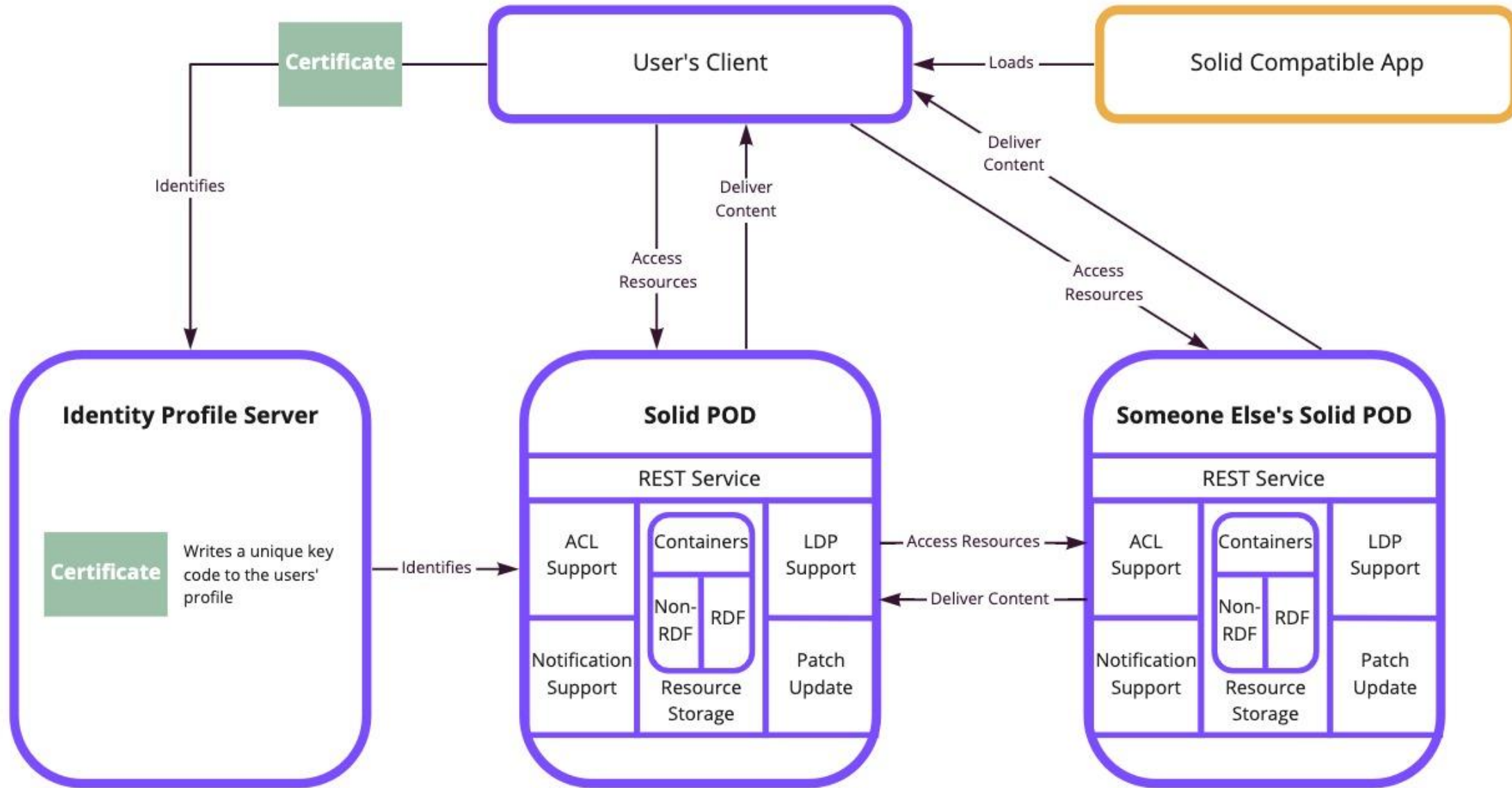
Scale laboratory  
accreditation & continual  
pipeline validation



Align economic incentives



# Solid's Architecture



miro

# Solid Pod Implementations?



BBC MyPDS App



Flemish Data Utility Company

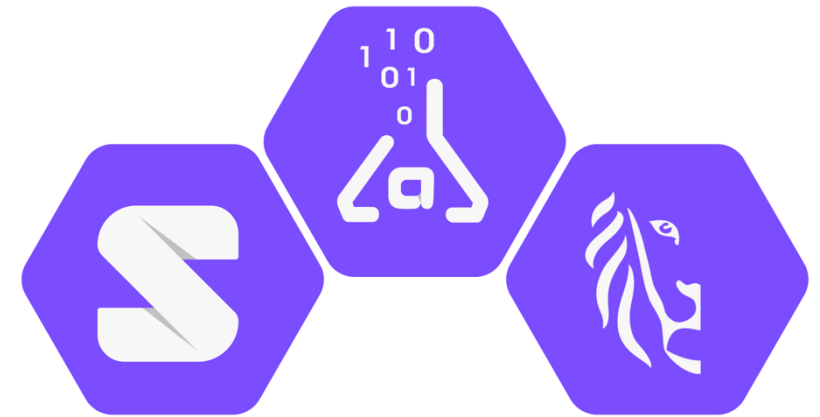


health data vaults



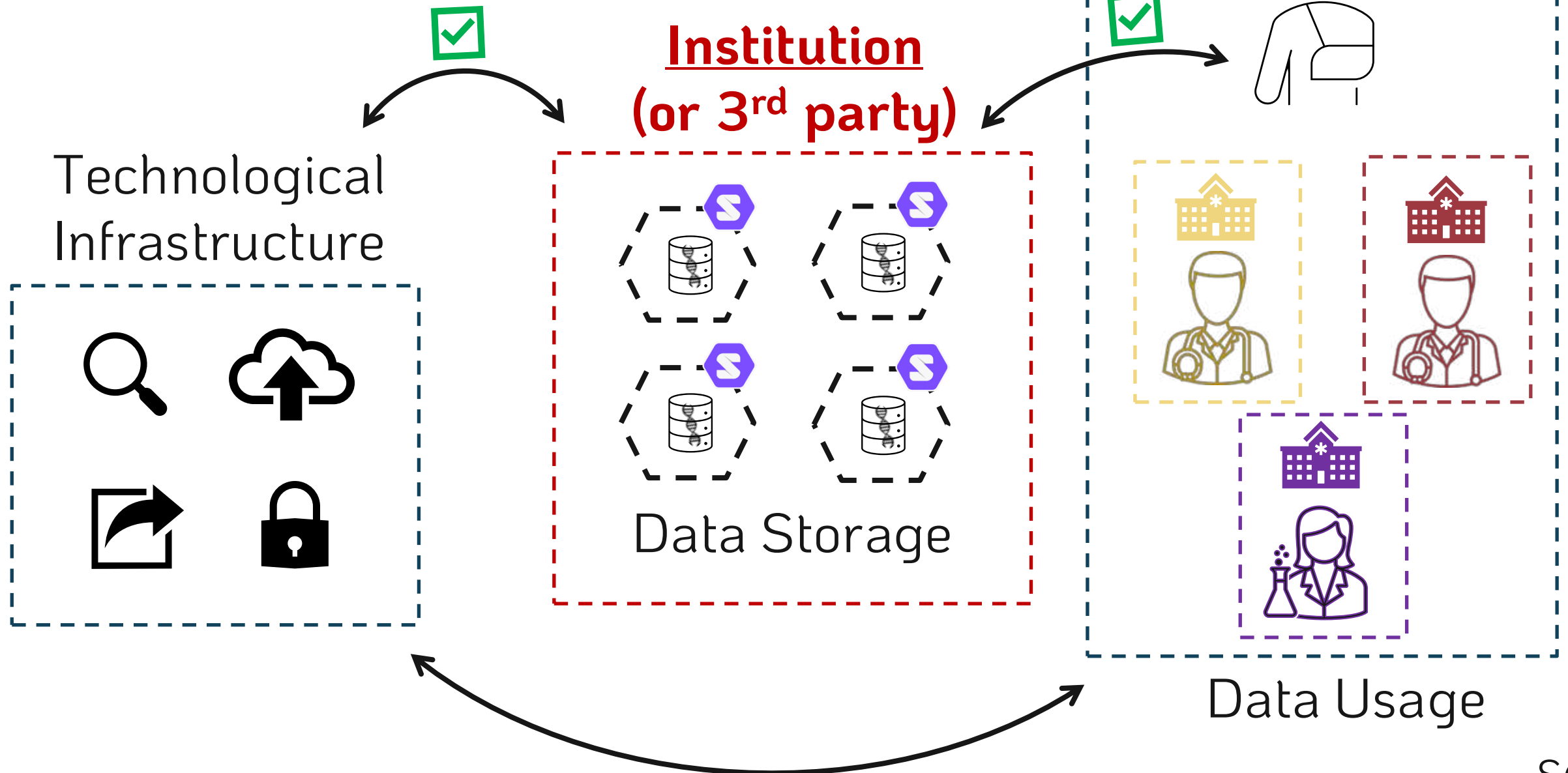
**Flanders**  
State of the Art

Flemish Smart Dataspace

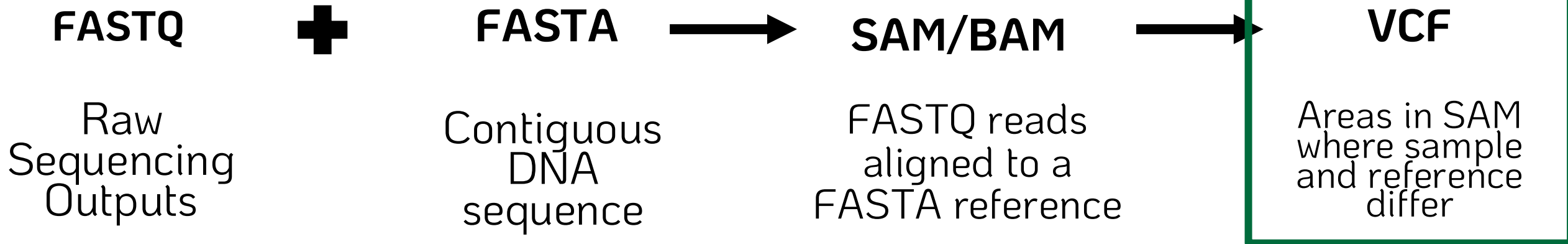


Solid Lab Use-cases

# A Production Solid Solution



## Semantic VCF – Sequencing Data Formats Intro



```

1 >gi|568336023|gb|CM000663.2| Homo sapiens chromosome 1, GRCh
2 CCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAAC
3 ACCCTAACCCCTAACCCCTAACCCCTAACCCCAACCCCTAACCCCTAACCCCTAAC
4 TAACCCCTAACCCCTAACCCCTAACCCCTAACCTAACCCCTAACCCCTAACCCCTA
5 CTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAAC
6 CAACCCCAACCCCAACCCCAACCCCAACCCCAACCCCTAACCCCTAACCCCTAAC
7 CCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAAC
8 TAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCG
9 TCTGACCTGAGGAGAAGTGTGCTCCGGCTTCAGAGTACCACCGC
10 CGCCCTCGCGGTGCTCTCCGGGTCTGTGCTGAGGAGAAACGCACA
11 CGGCGCAGGCGCAGAGAGGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCA
12 AGAGAGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCG
13 CGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCG
14 GCCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCG
15 AGGCGCACCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCG
16 GCAGAGACGCAAGCCTACGGCGGGGGTTGGGGGGGCGCGCGCGCGCGCGCGCG
17 GCTGGGGCGGGGGGAGGGTGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGAA
18 GGTAGAACCTCAGTAATCCGAAGCGCGGGATCGACCGCGCGCGCGCGCGCGCG
19 GCTTGCTCACGGTGCTGTGCCAGGGCGCCCCCTGCTGGCGACGCGCGCGCGCG
20 GAGTGGTGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCGCG

```

**@FORJUSP02**  
**CCGTCAATTC**  
**+**  
**AAAAAAAAAAAA**

**Q scores (as ASCII char)**  
**Base=T, Q=':'=25**

```

@HD VN:1.0 SO:0
@SQ SN:chr20
@PG ID:TopHat
lign-edit-dist 2 -
20 /data/user446/m
HWI-ST1145:74:C101D
CCGTGTTTAAA
C BBDCCDDCCDDDD
AS:i:-15
HWI-ST1145:74:C101D
TGCTGGATCAT
G DCDDDDDEDDDDDD
AS:i:-16
HWI-ST1145:74:C101DACXX:7:1204:14760:4030 16 chr20 270877 50 100M * 0 0
GGCTTTATTGGTAAAAAAGGAATAGCAGATTTAATCAGAAATCCCACCTGGCCAGCAGCACCAACCAGAAAGAAGGGAAGAAGACAGGAAAAACCA
C DDDDDDDDCDDDDDDDDDEEEEEEEFFFEFFEGHHHFGDJJTHJJJIJJJIIIGGFJJIIHIIIJJJJJJIGHHFAHGFIJHFGGHFFFDDBB
AS:i:-11 XM:i:2 XO:i:0 XG:i:0 MD:Z:0A85G13 NM:i:2 NH:i:1
HWI-ST1145:74:C101DACXX:7:1210:11167:8699 0 chr20 271218 50 50M4700N50M * 0
0 GTGGCTCTTCCACAGGAATGTTGAGGATGACATCCATGTCTGGGGTGCACTTGGGTCTCCAAGCAGAACATCTCAAATATGACCTCTCG
accepted hits.sam

```

# What does a VCF File Look like?

## Example

**VCF header**

```
##fileformat=VCFv4.0
##fileDate=20100707
##source=VCFtools
##reference=NCBI36
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality (phred score)">
##FORMAT=<ID=GL,Number=3,Type=Float,Description="Likelihoods for RR,RA,AA genotypes (R=ref,A=alt)">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##ALT=<ID=DEL,Description="Deletion">
##INFO=<ID=SVTYPE,Number=1,Type=String,Description="Type of structural variant">
##INFO=<ID=END,Number=1,Type=Integer,Description="End position of the variant">
```

**Mandatory header lines**

**Optional header lines** (meta-data about the annotations in the VCF body)

| #CHROM | POS | ID  | REF | ALT   | QUAL | FILTER | INFO               | FORMAT   | SAMPLE1  | SAMPLE2 |
|--------|-----|-----|-----|-------|------|--------|--------------------|----------|----------|---------|
| 1      | 1   | .   | ACG | A,AT  | .    | PASS   | .                  | GT:DP    | 1/2:13   | 0/0:29  |
| 1      | 2   | rs1 | C   | T,CT  | .    | PASS   | H2;AA=T            | GT:GQ    | 0 1:100  | 2/2:70  |
| 1      | 5   | .   | A   | G     | .    | PASS   | .                  | GT:GQ    | 1 0:77   | 1/1:95  |
| 1      | 100 | .   | T   | <DEL> | .    | PASS   | SVTYPE=DEL;END=300 | GT:GQ:DP | 1/1:12:3 | 0/0:20  |

**Body**

**Reference alleles** (GT=0)

**Alternate alleles** (GT>0 is an index to the ALT column)

**Deletion**

**SNP**

**Large SV**

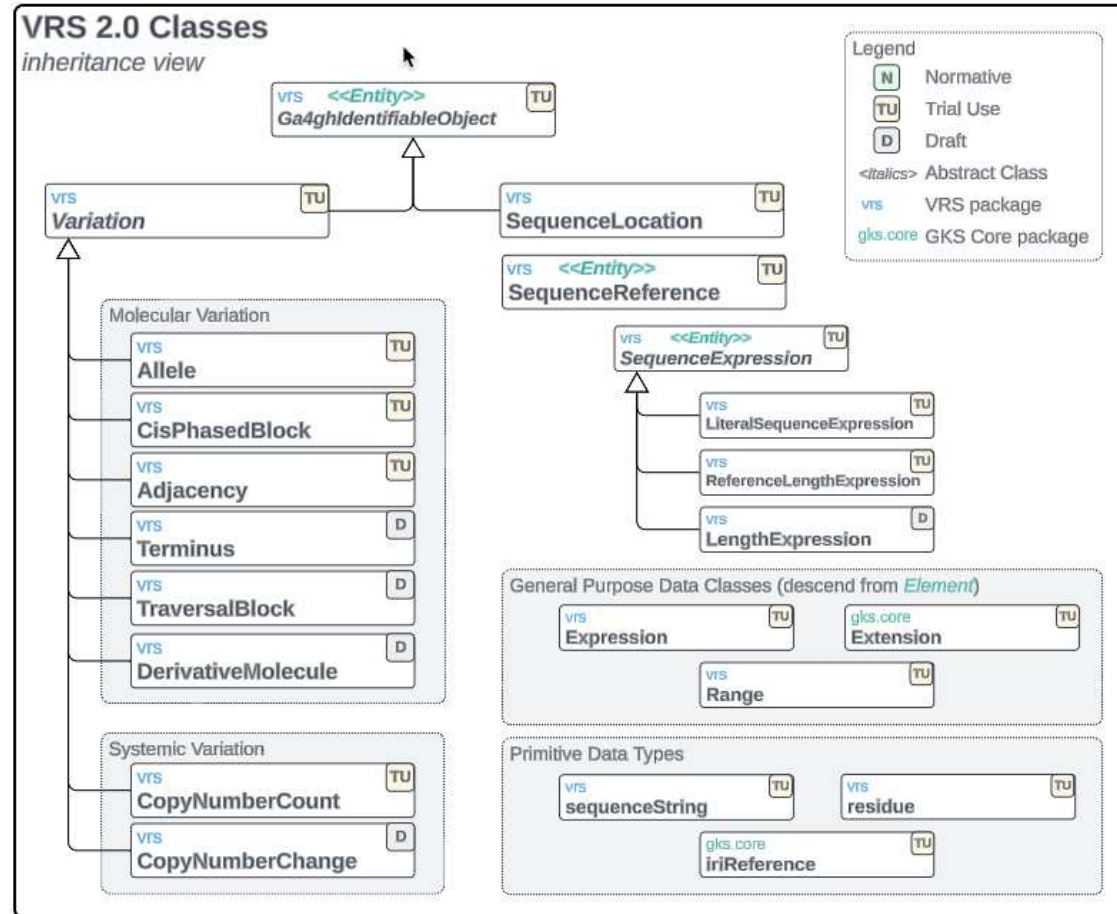
**Insertion**

**Other event**

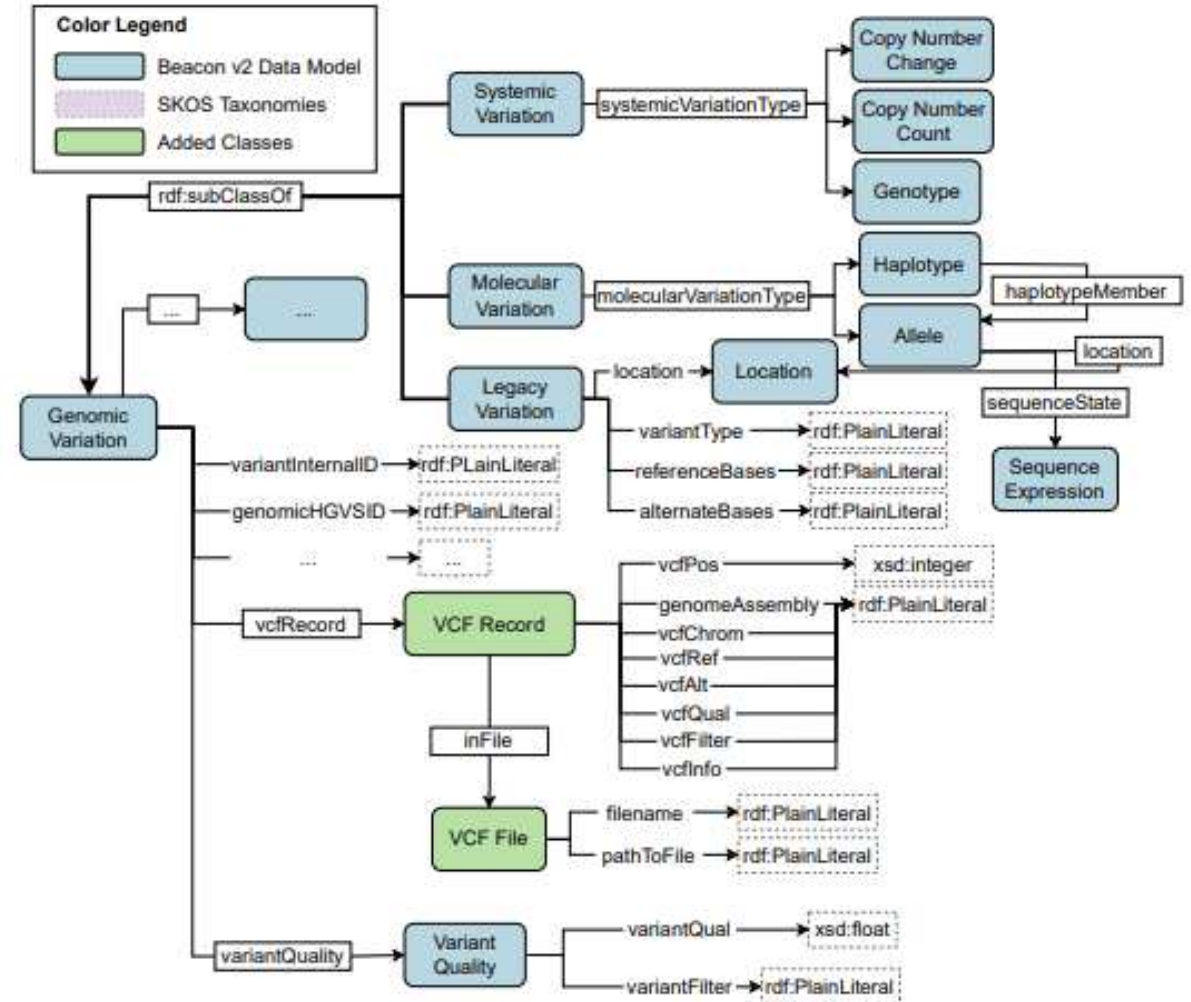
**Phased data** (G and C above are on the same chromosome)

# Representing VCF Semantically?

## GA4GH Variant Schema:



## HERO Genomics Ontology:

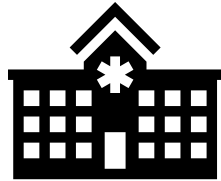
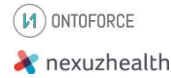


# Envisioned Product Evolution



PhD End

PhD  
Deliverable



Short Term

Start-up / Product  
development  
+  
Hospital implementation



Medical record



Medium Term

Integrate with EHR  
+  
Increase application  
domains



Long Term

Increase 3<sup>rd</sup> party  
application support  
+  
Scale to Europe and  
beyond